

G Chip 48

A chiptune sound module with five 8-bit and 16-bit era sound chips, tracker-style arps and built-in drum kits.



Overview

G Chip 48 is a chiptune synth built as a rack sound module. It recreates the sound chips of the 8-bit home computers and early games consoles: the one-bit BEEPER of the 48K home computers, the three-voice AY chip of the 128K machines with its buzzer envelope, the PSG square-wave chip with its white and periodic noise, the four-operator FM chip of the 16-bit consoles, and the PULSE of the 8-bit consoles with its 12.5, 25 and 50% pulses and stepped triangle. The sounds are designed from scratch.

Each chip behaves like the real thing: pitches snap to the chip's own tuning steps, volume snaps to its own level steps, and by default each chip has only as many voices as the original. The envelope, arpeggio, vibrato and pitch sweep all step once a frame (50 or 60 times a second), the way the old music drivers and trackers did it.

The rubber keys along the top pick the chip and the patch; the knobs below shape the sound. Each chip also has its own drum kit on the General MIDI drum notes.

Reach for it for

- Chiptune and video-game music
- Retro arpeggio chords, bleepy leads and buzzy basses
- Lasers, zaps and 8-bit drum kits

- 16-bit FM basses, brass and electric pianos
- Adding a lo-fi 8-bit layer to modern synth parts

Playing it

1. Create a track and add G Chip 48 as its effect: it is an instrument, so it has no audio input and produces stereo output.
2. Arm the track for recording with your MIDI keyboard as the input and monitoring on, or draw notes in a MIDI item on the track.
3. Pick a chip with keys 1 - 5, or click the screen for the full patch list. The MIDI IN light flashes as notes arrive.
4. By default it listens on all MIDI channels (OMNI). Click CH to choose a single channel if you drive several instruments from one source.

MIDI: pitch bend covers +/-2 semitones; the mod wheel (CC1) adds vibrato; the sustain pedal (CC64) holds notes; All Notes Off (CC123) and All Sound Off (CC120) stop everything.

Controls

Top left

CH (MIDI)	The MIDI channel it listens to: OMNI (all channels) or 1 - 16. Click to step up; Shift-click to step down.
TV (50 Hz / 60 Hz)	The frame rate: PAL (50 Hz) or NTSC (60 Hz). Everything that steps per frame - envelope, arp, vibrato, sweep and glide - runs at this rate, so NTSC is slightly faster.
MIDI IN	Lights when a note arrives.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

The screen

Shows the patch number and name, the chip, how many voices are playing out of how many are available, the frame rate, and ARP, HELD ARP or DRUMS when they are on. The border takes the chip's colour and runs loading stripes while notes play. Click the screen for the full patch list, grouped by chip.

The number keys

1 - 5 (chip)	1 BEEPER, 2 AY, 3 PSG, 4 FM, 5 PULSE. Pressing a key loads that chip's first patch. The lit key shows the current chip.
6 PATCH < PREV	The previous patch (wrapping round).
7 PATCH NEXT >	The next patch (wrapping round).
8 DRUMS	Switches the chip's drum kit on or off (see Drums below).
9 VOICES	CHIP keeps each chip's own voice count: BEEPER 1, AY and PSG 3, FM 6, PULSE 2 (1 when the mixer is on TRI). 6 lifts every chip to six voices. When all voices are busy, the oldest note is taken over.
0 PANIC	Stops every sounding note at once.

TONE / NOISE

DUTY	The pulse width, 5 to 50%. On the BEEPER the width also falls with the volume, which is how the one-bit speaker played loud and soft. On the PULSE it snaps to the chip's three widths: 12.5, 25 or 50%. The other chips always play a 50% square.
MIXER	TONE: the square or pulse. T+N: tone and noise combined, for a gritty, steamy sound. NOISE: noise only. TRI: a stepped 4-bit triangle, as on the PULSE.
NOISE	The noise pitch, 0 to 100 (70 by default). Higher is brighter and hissier.

TYPE

WHITE or PERIODIC noise, on the PSG and PULSE. On the PSG, periodic noise is a pitched buzz that follows the note; on the PULSE it is the chip's short-loop metallic noise.

BUZZ

AY only: the AY's envelope generator run at audio rate - its famous buzzer bass and lead sounds.

SHAPE

OFF, SAW, TRI or SYNC. SAW and TRI play the envelope as a waveform; SYNC gates it with the tone for a harder, synced sound.

TUNE

The buzz tuned an octave down, in unison or an octave up from the note. Its period also snaps to the chip's own steps.

ENVELOPE

Stepped once a frame, as on the chips.

ATTACK

0 to 2000 ms. 0 starts the note at full level.

DECAY

0 to 5000 ms (400 ms by default): the fall from full level to the sustain level.

SUSTAIN

0 to 15, the chips' own volume steps (12 by default). At 0 the note dies away after the decay, for plucks and bells.

RELEASE

0 to 5000 ms (120 ms by default): the fade after you let go.

ARP

CHORD

OFF, MAJ, MIN, OCT, 5TH, MAJ7 or MIN7: each note becomes a fast tracker-style arpeggio of that chord on a single voice. HELD steps through every note you hold, in the order you played them, on one voice.

SPEED

How many frames each arp step lasts, 1 to 8 (1 is the fastest).

VIBRATO

DEPTH	0 to 100 cents.
DELAY	How long after the note starts the vibrato comes in, 0 to 1000 ms (250 ms by default).
RATE	1 to 12 Hz (6 Hz by default).

PITCH

GLIDE	Slides from the last note to the new one, 0 to 1000 ms. 0 is off.
SWEEP	Each note starts this far away from its pitch, -48 to +48 semitones, and sweeps back to it: lasers, zaps and drum-like drops.
TIME	How long the sweep takes, 10 to 2000 ms (150 ms by default).

FM

FM only, except DETUNE / PHASER. The operator tunings and levels come from the FM patch you load; these knobs shape them.

ALGO	Which of the eight four-operator algorithms is used, 0 to 7: from all four operators in a stack (the most complex tones) to all four as carriers (organ-like).
FEEDBACK	Operator 1 feeding back on itself, 0 to 7. Higher values add buzz and noise.
MOD	The overall modulation level, 0 to 200% (100% by default). More is brighter.
MOD DECAY	Scales how long the modulators take to die away, 10 to 400%. Short gives plucky, percussive tones; long keeps them bright.
DETUNE / PHASER	On FM, detunes the operators slightly against each other, 0 to 100%. On the BEEPER the knob reads PHASER and

	mixes a second, slightly detuned square, for the one-bit phasing sound.
LADDER	The grit of the early FM chip's converter, 0 to 100% (50% by default): coarser steps and a little step at the zero crossing.

OUTPUT

VELO	How much velocity changes the level, 0 to 100% (30% by default). 0 plays every note at full level.
LO-FI	Blends in the original machine's own speaker or output filter, 0 to 100% (60% by default): the BEEPER's little speaker (280 Hz to 4.2 kHz), the FM console's early output filter (rolled off from 3.4 kHz), and gentler filtering on the others.
STEREO	Spreads the voices across the stereo field, 0 to 100% (40% by default). 0 is mono.
VOLUME	The output level, -24 to +6 dB. Each patch is already level-matched, so they play at about the same loudness.

There is no input or output fold-out on this panel. The output passes through the GFX soft clipper, which catches peaks at -0.3 dBFS.

Drums

With DRUMS on, each chip plays its own kit on the General MIDI drum notes: 35 / 36 kick, 37 - 40 snare, 42 / 44 closed hi-hat, 46 open hi-hat, 49 / 52 / 55 / 57 crash, 51 / 53 / 59 ride. Toms (41 - 50) and every other note play pitched toms. The drums use six voices and ignore the envelope, arp and vibrato.

Presets

The patches set the chip and every sound control (tone, buzz, envelope, arp, vibrato, pitch, FM and drums). MIDI channel, frame rate, voices, velocity, lo-fi, stereo and volume are left as they are.

BEEPER

48K Lead	A plain square lead with light vibrato.
Thin Pulse	A thin 10% pulse lead.
Phaser Lead	The one-bit phaser sound (PHASER at 30%).
Beeper Bass	A short 30% pulse bass with a little glide.
Chord Arp	HELD arp: play a chord and it cycles through the notes.
Laser Zap	A 36-semitone sweep down onto the note with no sustain.
Beeper Drums	The beeper drum kit.
Soft Hum	A soft 25% pulse with a slow attack, long release and gentle vibrato.

AY

AY Lead	A square lead with vibrato.
Buzzer Bass	The saw-shaped buzzer in unison: the classic AY bass.
Buzz Tri Bass	The buzzer with a triangle shape, softer.
Sync Lead	The buzzer synced to the tone, an octave up, with vibrato.
Arp Chords	HELD arp chords.
Soft Bell	A plucked square that dies away, with a little vibrato.
Steam Lead	Tone and noise combined for a breathy lead.
AY Drums	The AY drum kit.

PSG

PSG Lead	A square lead with vibrato.
PSG Bass	A short square bass.
Periodic Bass	Periodic noise played as a pitched, buzzy bass.

Major Arp	A major-chord arp, 2 frames a step.
Minor Arp	A minor-chord arp, 2 frames a step.
PSG Bell	A plucked square that dies away.
Vib Square	A square with deep, immediate vibrato.
PSG Drums	The PSG drum kit.

FM

16-Bit Bass	A punchy FM bass with extra feedback and ladder grit.
FM Brass	Brass with a short attack, vibrato and a little detune.
FM E.Piano	An electric piano with a long decay and no sustain.
FM Lead	A bright lead with strong feedback and vibrato.
FM Strings	Slow attack and release with plenty of detune.
FM Marimba	A short, woody mallet sound.
FM Organ	All four operators as carriers, full sustain.
FM Drums	The FM drum kit.

PULSE

Pulse 12.5	The thinnest of the chip's pulses.
Pulse 25 Lead	The 25% pulse with vibrato.
Square 50	The plain 50% square.
Triangle Bass	The stepped triangle as a bass (one voice).
Echo Arp	A 25% pulse major-chord arp.
Vibrato Lead	A 25% pulse with deep vibrato.

Slide Lead	A square lead with glide between notes.
8-Bit Drums	The PULSE drum kit.

Init

Init	A plain BEEPER square with a basic envelope: a starting point for your own sounds.
-------------	--

Tips

- For authentic chip chords, use an ARP chord or HELD rather than playing chords with more voices than the chip has.
- Keep VOICES on CHIP for the real thing: note stealing on a one-voice BEEPER is part of the sound.
- Lasers and drum drops: set SUSTAIN to 0 and use a large SWEEP with a short TIME.
- Turn LO-FI down for a cleaner, fuller sound in a modern mix; up for the original small-speaker character.

File: gfx-g-chip.jsfx